NLP MID-PROJECT SUBMISSION-REPORT

TOXCL (Hoang et al., 2024) — "A Unified Framework for Toxic Speech Detection and Explanation"

Problem Statement:

Vastly hate detection systems focus on identifying whether a sentence is toxic or not, but struggle to understand and justify the implicit hate. Such hate is often expressed through sarcasm, stereotypes, or coded language, where the harmful meaning is not directly stated. As shown in prior work, models trained on explicit hate fail to capture these hidden meanings and often produce incorrect or shallow explanations.

Recent approaches try to generate explanations (e.g., implied stereotypes or target groups), but these models either produce generic outputs or rely too heavily on surface patterns instead of true reasoning. Even knowledge-enhanced models improve performance only partially, as they still fail to handle context, ambiguity, and complex linguistic structures.

Therefore, there is a need for a model that can not only detect toxic language but also accurately recover the implied meaning behind it, while being robust, interpretable, and efficient. This is the problem that TOXCL aims to address by combining transformer-based architectures with improved reasoning and knowledge transfer mechanisms.

Literature Survey

Foundational Benchmarks & Systematic Surveys:

The development of TOXCL builds on the shift in NLP from simple hate speech classification to deeper reasoning-based understanding. Early survey work shows that traditional methods like SVM and TF-IDF struggled with context and ambiguity, while transformer models such as BERT improved performance by capturing semantic meaning. However, key challenges remain, including handling sarcasm, ambiguity, and poor generalisation across datasets.

A major step forward came with *Latent Hatred* (ElSherief et al., 2021), which highlighted that implicit hate, expressed through sarcasm, stereotypes, and coded language, is much harder to detect than explicit hate. The Implicit Hate Corpus introduced in this work focuses on recovering the **implied meaning** behind text, which is central to TOXCL's objective.

Similarly, *Social Bias Frames* (Sap et al., 2020) reframed toxicity detection as a reasoning task. Instead of predicting a single label, it models multiple aspects such as **target group, intent, and implied stereotype**. This work motivates the use of generation models (like T5 in TOXCL), but also shows that such models often produce vague or generic explanations.

To address this, Sridhar and Yang (2022) propose a knowledge-enhanced generation framework (MIXGEN) based on BART. Their approach combines expert annotations, knowledge graphs, and pretrained language models to generate better explanations. While this improves performance, the model still struggles with deeper reasoning, especially in cases involving sarcasm or complex context.

Further work by Ocampo et al. (2023) shows that implicit and subtle hate require different modelling approaches, and that models still perform poorly on such cases despite high accuracy on explicit hate. This highlights that the main challenge is understanding hidden meaning rather than improving model size alone.

From an architectural perspective, Raza et al. (2024) demonstrate that combining multiple models (transformers, CNNs, RNNs) improves robustness, but models remain sensitive to adversarial changes like paraphrasing. This reinforces the need to combine architectural improvements with better reasoning and generalisation.

Overall, these works show that effective toxicity modelling requires moving beyond classification toward explanation, integrating external knowledge, and handling implicit meaning. TOXCL builds on these ideas by using a transformer-based architecture with knowledge integration and distillation to produce more accurate and interpretable explanations.

References (with links)

1. Jahan & Oussalah (2023) — A systematic review of Hate Speech automatic detection using Natural Language Processing
2. ElSherief et al. (2021) —Latent Hatred: A Benchmark for Understanding Implicit Hate Speech
3. Sap et al. (2020) — SOCIAL BIAS FRAMES: Reasoning about Social and Power Implications of Language
4. Sridhar & Yang (2022) — Explaining Toxic Text via Knowledge Enhanced Text Generation
5. Ocampo et al. (2023) —An In-depth Analysis of Implicit and Subtle Hate Speech Messages
6. Raza et al. (2024) — Hate speech detection on Twitter using transfer learning

Methodological Core (Architecture & Distillation)

The architecture of generative toxicity detection has historically struggled with the conflicting optimisation trajectories of discriminative classification and conditional explanation generation. Frameworks often mix these objectives, leading to severe error propagation and exposure bias. The ToXCL framework resolves this through a tripartite modular architecture that bridges the gap between instruction-tuned sequence-to-sequence backbones and discriminative knowledge distillation.

ToXCL leverages the Flan-T5 architecture. Prior work demonstrated that scaling instruction-finetuned language models significantly enhances task generalisation across heterogeneous formats (Chung et al., 2022). This provides an ideal backbone for ToXCL, enabling the encoder to process bidirectional context for implicit toxicity detection while the decoder autoregressively generates explanatory rationales. However, relying on a unified generation sequence for complex reasoning introduces compounding errors.

AlKhamissi et al. (2022) established via the ToKen framework that explicitly decomposing hate speech detection into logical subtasks, specifically generating the implied target group before final classification, vastly improves discriminative performance. Building on this theoretical necessity while mitigating autoregressive exposure bias, ToXCL structurally decouples this reasoning step into an independent Target Group Generator. This isolation allows the primary classifier to evaluate text with a fully realised demographic context without sequential error cascading.

To robust the classification capabilities of the generative Flan-T5 encoder without inflating inference latency, ToXCL integrates a Teacher Classifier via knowledge distillation (KD). Originating from Hinton et al. (2015), response-based KD leverages temperature-scaled soft targets to transfer the high-dimensional decision boundaries of a complex model to a more compact student. As KD taxonomies expanded to include intermediate feature representations (Gou et al., 2021), mitigating semantic mismatch across heterogeneous neural architectures became critical. Methodologies like Semantic Calibration (SemCKD) by Chen et al. (2021) utilise attention mechanisms to dynamically align disparate layer semantics, preventing negative regularisation. This literature enables ToXCL's ability to safely map the highly calibrated parameters of an encoder-only teacher onto the hybrid Flan-T5 student encoder. Furthermore, recent advances in multi-student mutual learning (DDML, 2024) validate that such distillation paradigms act as crucial topological regularizers for highly subjective, out-of-distribution hate speech.

By grounding an instruction-tuned generative decoder on a decoupled demographic generator and a knowledge-distillation classification encoder, ToXCL establishes a highly deployable, mathematically stable foundation for modular toxicity detection.

References with links

1. Badr AlKhamissi-ToKen: Task Decomposition and Knowledge Infusion for Few-Shot Hate Speech Detection
2. Cross-Layer Distillation with Semantic Calibration
3. Scaling Instruction-Finetuned Language Models
4. Knowledge Distillation: A Survey
5. Distilling the Knowledge in a Neural Network
6. ToXCL: A Unified Framework for Toxic Speech Detection and Explanation
7. DDML: Multi-Student Knowledge Distillation for Hate Speech.

Recent Advances & LLM Frontiers (2023–2024)

Between 2023 and 2024, the detection of toxic speech moved away from simple "toxic vs. safe" labels toward models that can explain their reasoning. This shift was led by Large Language Models (LLMs) that act as "generative" thinkers rather than just classifiers. **Mistral 7B (Jiang et al., 2023)** set a new standard here, showing that even smaller, efficient models could detect hate speech with high accuracy without needing massive amounts of specific training data. This created a strong baseline for others to follow.

However, a major problem with these models is that they sometimes give a correct "toxic" label but provide a confusing or unrelated explanation. **TOXCL (Hoang et al., 2024)** fixed this by introducing a "Conditional Decoding Constraint." This technical rule ensures the model's explanation is strictly tied to its final label, preventing the AI from making up illogical reasons. Research is also becoming more precise about where the "hate" is located. **Jafari et al. (2024)** moved beyond looking at whole sentences to find the exact words, or "linguistic triggers," that cause harm in implicit hate speech.

The latest research also handles more than just text. **Huang et al. (2024)** and **Hee et al. (2024)** showed how AI can now detect hate in memes and photos by connecting visual clues with captions. This is vital because modern hate speech is often hidden in images. Finally, researchers like **Masud et al. (2024)** and **Antoniak et al. (2024)** warn that AI often learns human biases from training data and can miss hate when it is hidden in long stories. Together, these 2024 breakthroughs are moving the field toward AI that understands the complex, human ways that harmful content is shared.

References with links

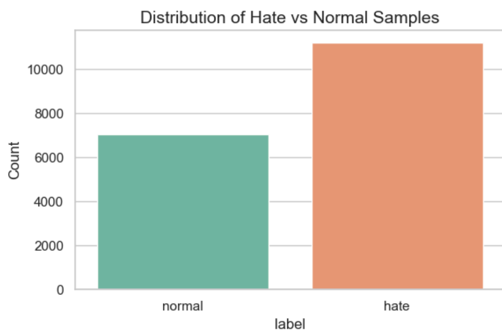
1. TOXCL (Hoang et al., 2024) — ToXCL: A Unified Framework for Toxic Speech Detection and Explanation
2. Jafari et al. (2024) — Implicit Hate Target Span Detection in Zero- and Few-Shot Settings with Selective Sub-Billion Parameter Models
3. Huang et al. (2024) — "Towards Low-Resource Harmful Meme Detection with LMM Agents
4. Masud et al. (2024) — Human and LLM Biases in Hate Speech Annotations: A Socio-Demographic Analysis of Annotators and Targets
5. Antoniak et al. (2024) — Where Do People Tell Stories Online? Story Detection Across Online Communities
6. Hee et al. (2024) — Bridging Modalities: Enhancing Cross-Modality Hate Speech Detection with Few-Shot In-Context Learning

Description of the dataset used in the Paper(Code for graph generation already in codebase)

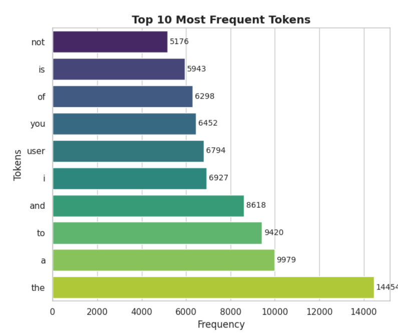
Hate Xplain Dataset :

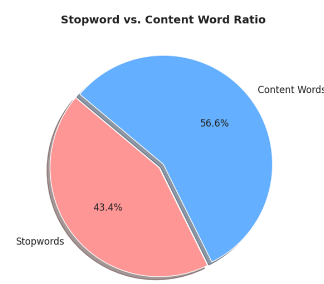
1. Problem Formulation and Context This dataset represents a dual-objective Natural Language Processing task: **Hate Speech Classification** combined with **Targeted Demographic Identification**. The objective is not merely to detect toxicity, but to pinpoint the specific community or identity group being attacked.

In the context of the *ToxCL (Toxic Speech Detection and Explanation)* framework, this dataset serves as the foundational training corpus for the Target Group Generator (TG) module. By teaching the model to explicitly identify target demographics, the framework forces the architecture to learn contextual markers rather than relying on generic, superficial toxicity features. This grounds the subsequent encoder-decoder modules, enabling them to generate accurate, human-readable explanations of *implicit* hate.



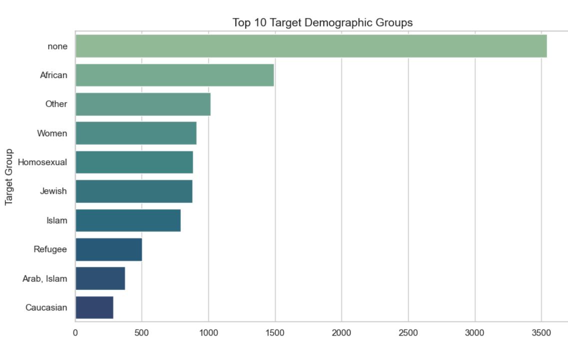
2. Dataset Overview and Distribution To train the generator and facilitate demographic aware toxicity detection, we analyze 18,224 instances. The dataset exhibits a pronounced class imbalance favoring the ***`hate`*** label (61.4% hate vs. 38.6% normal).





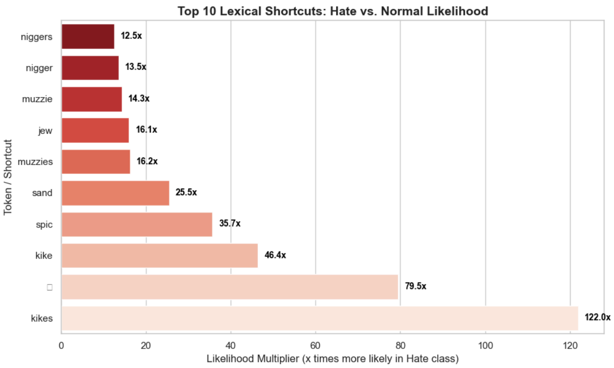
3. Linguistic Characteristics The textual instances exhibit an average length of 23.6 tokens. Dataset draws from a highly diverse vocabulary of approximately 27,000 unique unigrams. This inflated vocabulary size reflects the colloquialisms, deliberate misspellings, and non-standard literature words typical of raw social media streams. The frequent presence of anonymised tokens (e.g., ***`<user>`***) indicates that direct personal identifiers and mentions were scrubbed during collection, shifting the linguistic focus entirely onto the semantic content of the posts.

4. Intersectional Target Demographics A critical analysis of the annotation taxonomy reveals a heavily intersectional distribution of hostility.



Attacks are prominently directed toward African (1,494 samples), Women (913), Homosexual (885), and Jewish (879) communities. Neutral demographic identifiers (e.g., *white, women, black*) appear with extreme frequency within the corpus. This overlap may introduce a bias, if a classifier encounters demographic markers strictly within toxic contexts, it risks inherently associating neutral identities with toxicity and heavily penalize non toxic discussions regarding minority groups.

5. Toxicity Indicators and Lexical Shortcuts: A critical flaw observed in this explicit toxicity dataset is the deep embedding of annotation artifacts and lexical shortcuts. A shortcut occurs when a specific token correlates so heavily with a label that a classifier memorizes the token rather than learning the underlying semantics.

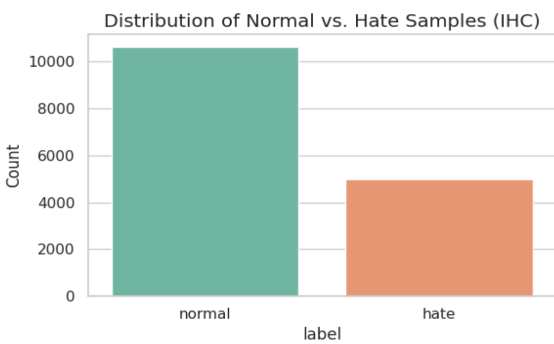


Our statistical odds-ratio testing demonstrates that specific slurs and emojis are disproportionately confined to the toxic class. As illustrated in Figure 4, tokens such as ***`kikes`*** (122x more likely in the hate class), the vomiting emoji 🤮 (79.5x likelihood -> second last entry in graph), and ***`spic`*** (35.6x likelihood) act as massive shortcuts. While these unigrams serve as highly accurate signals for baseline explicit classification, they act as devastating shortcuts for deep learning architectures.

6. Impact on Model Training If an unconstrained model encounters these shortcut tokens, it bypasses linguistic reasoning and defaults to a ***`hate prediction`***. Without advanced mitigation strategies such as the knowledge distillation models may memorise these shortcuts rather than understanding the semantics. Hence, that's why ToxCl only uses it to generate the target group rather than training and evaluating the ToxCl framework on this data.

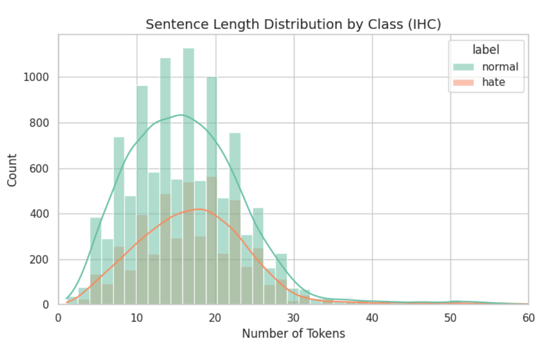
IHC Dataset

1. Problem Formulation and Implicit Hate Unlike datasets that focus on explicit toxicity (characterised by slurs and profanity), this dataset represents the **Implicit Hate Corpus (IHC)**. Implicit hate utilizes coded language, stereotypes, and ostensibly polite framing to attack demographic groups, making it notoriously difficult for standard NLP architectures to detect. The objective of this dataset is not only to classify a text as ***`hate`*** or ***`normal`***, but to simultaneously extract the ***`Target`*** demographic and provide a generative ***`explanation`*** (e.g.,*"Immigrants should be deported"*).

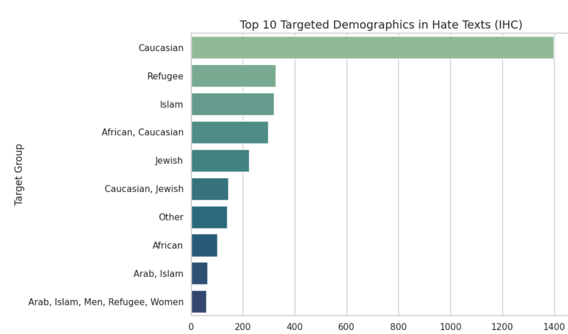


2. Dataset Overview and Distribution The IHC train corpus consists of 15,635 samples. In stark contrast to explicit hate datasets, this dataset exhibits a class imbalance that heavily favors the ***`normal`*** label (68.0% normal vs. 32.0% hate).

This "normal-heavy" distribution reflects the realistic distribution of social media channels and prevents models from defaulting to over-aggressive flagging. The sentences are relatively brief, averaging 16.7 tokens.Despite their brevity, these texts demand deep semantic comprehension, as the toxicity relies entirely on the contextual arrangement of standard vocabulary rather than the presence of lengthy, aggressive rants.



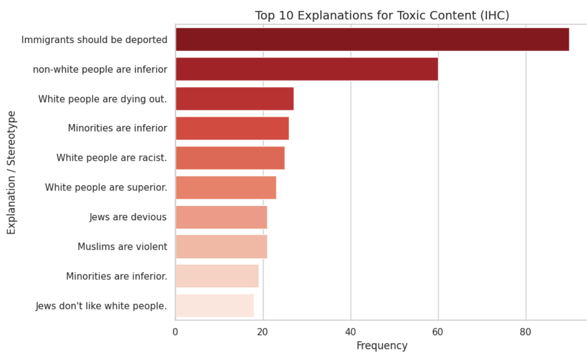
3. Targeted Demographics and White Supremacist Discourse Extracting the embedded target markers reveals a unique



demographic profile compared to standard explicit datasets.

The highest frequency of targeted discourse involves the ***Caucasian*** demographic (1,398 samples). However, a qualitative analysis of the corresponding texts reveals that these instances largely comprise alt-right and white nationalist talking points (e.g., *"White genocide," "White people are dying out"*). Other highly targeted groups include ***Refugee*** (327 instances), ***Islam*** (319 instances), and ***Jewish*** (224 instances). The dataset captures intersectionality heavily, frequently linking targets such as ***African***, ***Caucasian*** to draw implicit, racist comparisons.

4. Stereotype Explanations and Rationale A defining feature of the IHC dataset is the inclusion of ground-truth explanations detailing *why* the implicit text is toxic.



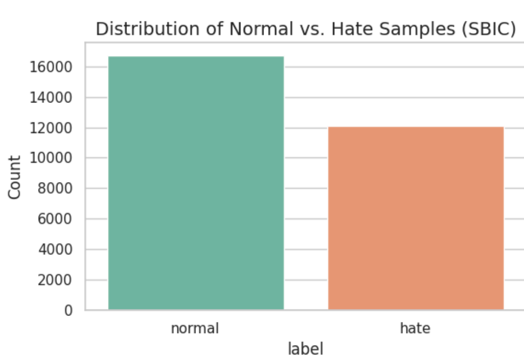
The most frequent stereotype explanations include "Immigrants should be deported," "non-white people are inferior," and "White people are dying out." Providing these exact rationales allows a framework like ToxCL to train encoder-decoder structures to decode the underlying indicators of hate and output the specific hatred being invoked.

5. The Absence of Lexical Shortcuts The most critical linguistic finding in this corpus is the **near-total absence of lexical shortcuts**. In our prior analysis of explicit toxicity, slur unigrams functioned as massive shortcuts (e.g., appearing 120x more often in the hate class). However, in the IHC dataset, our odds-ratio statistical testing yielded almost no high-impact shortcuts.

The only notable, yet vastly weaker, correlating tokens were ***deport*** (13.6x more likely in the hate class) and ***illegals*** (10.2x likelihood). The overall vocabulary is dominated by seemingly neutral words (*white, people, race, black*). Because a classifier cannot simply memorize a dictionary of profanity to achieve high accuracy, this dataset effectively enforces semantic reasoning, proving its essential value for training advanced, context-aware NLP models.

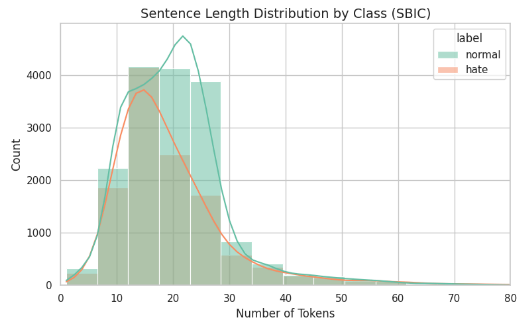
SBIC Dataset

1 Problem Formulation and Domain Dynamics Unlike traditional explicit hate speech corpora, the Social Bias Inference Corpus (SBIC) is uniquely tailored toward detecting and explaining microaggressions, harmful tropes, and offensive framing wrapped in dark humour. The classification task mirrors the dual objective of the ToxCL framework: assigning a binary toxicity label while simultaneously extracting the ***Target*** demographic and generating a human-readable ***Explanation*** of the underlying social bias.

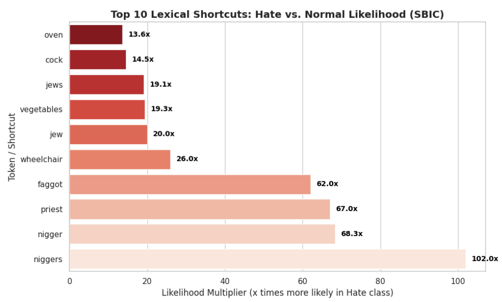


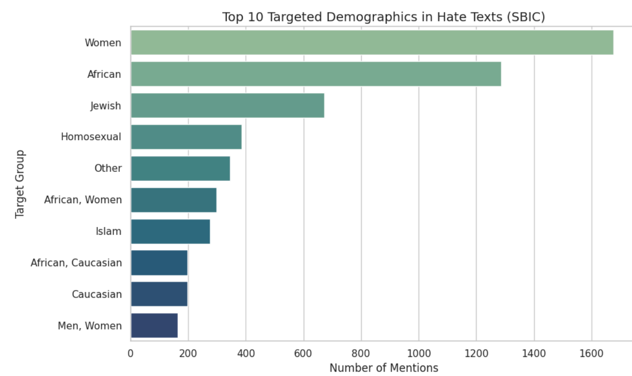
2 Dataset Overview and Distributions The SBIC training set comprises an expansive 28,796 instances. The dataset exhibits a moderate class balance, favouring the ***normal*** label (58.0% normal vs. 42.0% hate), ensuring a robust baseline for preventing false positives in non-toxic social media conversations. Textual sequences are highly concise, with the vast majority falling between 10 to 25 tokens. This necessitates that the encoder architecture should extract deep structural meaning from limited context.(graph on next page)

3 Intersectional Demographics and Generative Stereotyping A macro-level analysis of the `Target` identifiers reveals a distinct departure from political and alt-right talking points observed in other corpora. The most heavily targeted demographics within the SBIC toxic class are Women (1,675 samples), African communities (1,287), Jewish populations (672), and Homosexual individuals (386). Furthermore, the dataset contains a high volume of intersectional attacks (e.g., targeting *African Women* simultaneously).

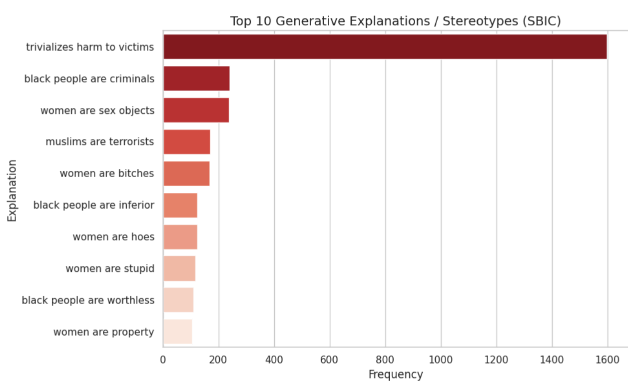


The novelty of the SBIC corpus lies in its ground-truth generative explanations. Statistical aggregation of these rationales highlights structural societal prejudices rather than generic anger. The dominant explanation across the dataset is the overarching phrase *"trivializes harm to victims"* (occurring over 1,500 times), functioning as a meta-label for rape culture and assault apologia. Additionally, explicit structural stereotypes such as *"women are sex objects," "black people are criminals,"* and *"muslims are terrorists"* dominate the explanatory distribution. Providing these highly specific rationales allows advanced generative architectures to learn the mapping between a seeming "joke" and its inherently harmful meaning.





4 The Resurgence of Dark Humor Lexical Shortcuts Interestingly SBIC dataset is the nature of its lexical shortcuts. While the Implicit Hate Corpus (IHC) was largely devoid of shortcuts, the dark humor and "edgy meme" nature of SBIC has generated highly specific, contextual shortcuts.

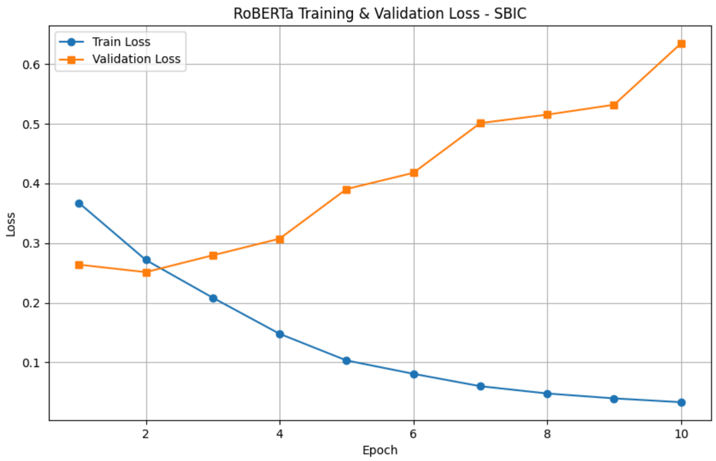


Odds-ratio evaluation reveals that alongside explicit slurs (e.g., ***niggers*** appearing 102x more frequently in the toxic class), neutral words have been co-opted into devastatingly predictive shortcuts due to meme-culture stereotyping. For example, the tokens ***wheelchair*** (26x likelihood) and ***vegetables*** (19.3x likelihood) act as massive shortcuts for attacks targeting paralysis/coma patients. Similarly, the token ***oven*** (13.6x likelihood) acts as a high-signal shortcut for antisemitic Holocaust "humour," while the ***priest*** (67x) correlates intensely with jokes regarding pedophilia. If models are deployed without careful demographic decoupling and knowledge distillation (as proposed in ToxCL), they risk intrinsically mapping these neutral nouns like *wheelchair* or *oven* to toxicity, thereby penalising innocent conversations.

Results of the baseline model

G1- Implicit toxic speech detection with classification only. (Roberta Base)

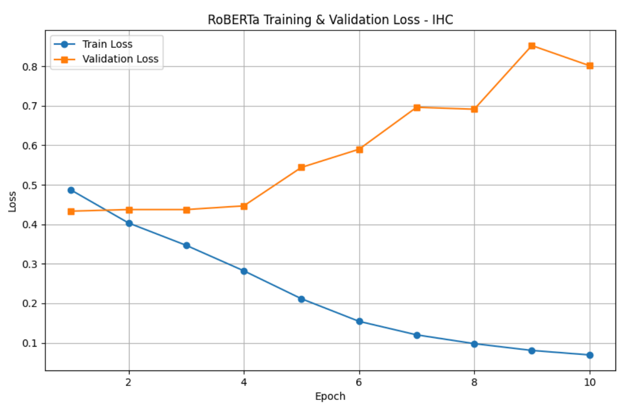
<u>SBIC Dataset tested on the test set</u>




```
{
    "Accuracy": 90.16645326504481,
    "Macro F1": 90.16464150842411
}
```


The training and validation plots of finetuned Roberta Base on the SBIC dataset are given. The metrics on the IHC are also reported.

<u>IHC Dataset(tested on validation set)</u>

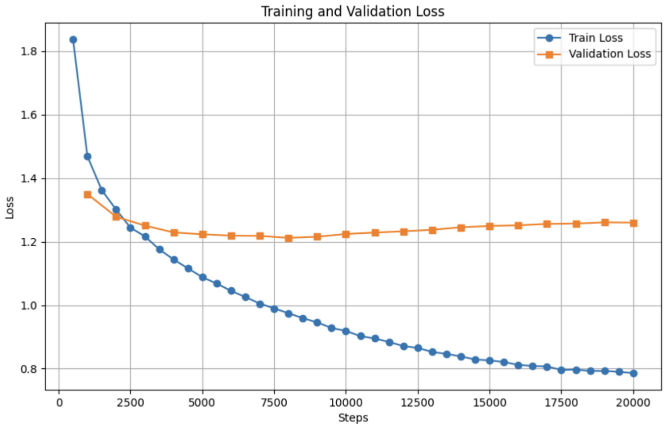



```
{
    "Accuracy": 79.95910020449898,
    "Macro F1": 77.4969646294588
}
```


The training and validation plots of finetuned Roberta Base on the IHC dataset are given. The metrics on the IHC are also reported.

G2- Implicit toxic speech explanation (BART)

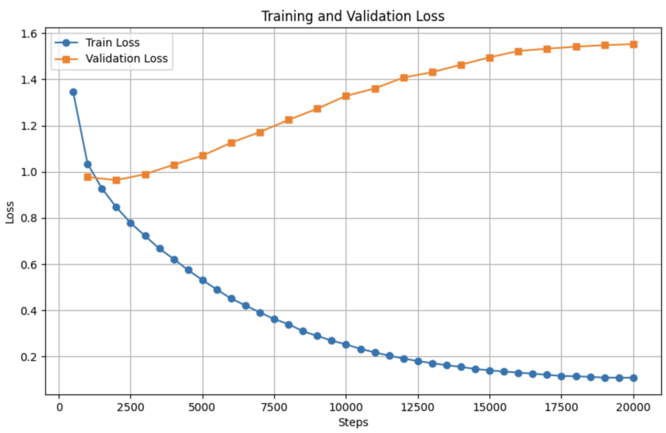
<u>SBIC dataset (tested on test set)</u>




```
G2-BART – SBIC
======================
BLEU-4    : 47.6493
ROUGE-L   : 69.5413
METEOR    : 66.4507
BERTScore : 84.0726
```

The training and validation plots of finetuned BART on SBIC dataset is given. The metrics on the SBIC are also reported.

<u>IHC dataset (validation results)</u>

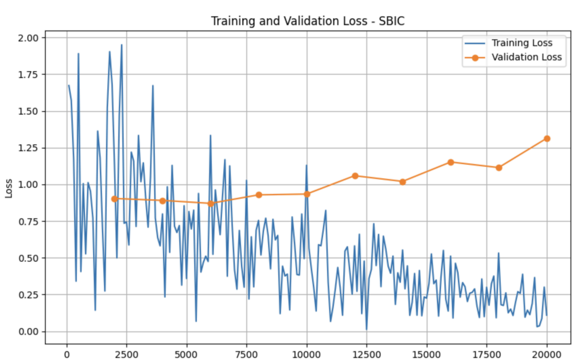



```
G2-BART – IHC
======================
BLEU-4    : 59.7603
ROUGE-L   : 66.7260
METEOR    : 65.3093
BERTScore : 77.1108
```

The training and validation plots of finetuned BART on IHC dataset is given. The metrics on the test IHC are also reported.

G3- Implicit toxic speech detection and explanation (GPT-2)

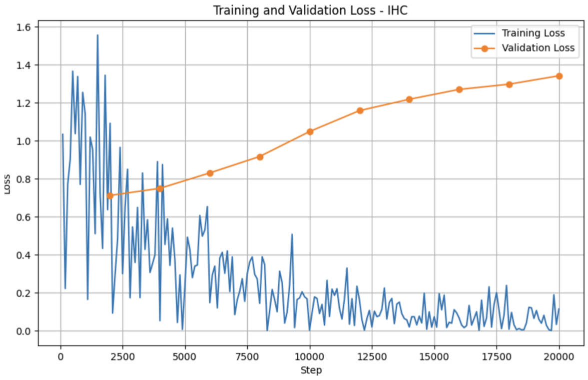
<u>SBIC dataset (tested on test set)</u>




```
{
    "Accuracy": 87.58002560819462,
    "Macro F1": 87.56538362474755,
    "BLEU-4": 45.57959217858083,
    "ROUGE-L": 66.5451515230672,
    "METEOR": 63.459667093469385,
    "BERTScore": 83.42195440040813
}
```

The training and validation plots of finetuned GPT-2 on the SBIC dataset are given. The metrics are also reported.

<u>IHC dataset(tested on validation set)</u>



"Accuracy": 77.5562372188139,
"Macro F1": 72.61303671809591,
"BLEU-4": 60.167157051706724,
"ROUGE-L": 66.15163191476357,
"METEOR": 64.82617586912033,
"BERTScore": 76.19880567055294

The training and validation plots of finetuned GPT-2 on the IHC dataset are given.The metrics are also reported..